



5.2 Euler's Method

Formula

$$w_0 = \alpha,$$

$$w_{i+1} = w_i + hf(t_i, w_i), \quad \text{for each } i = 0, 1, \dots, N-1.$$

$$w_{i+1} = w_i + hf(t_i, w_i)$$

Illustration In Example 1, we will use an algorithm for Euler's method to approximate the solution to

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5,$$

$\underbrace{\hspace{1.5cm}}_{f(t_i, w_i)} \quad \underbrace{\hspace{1.5cm}}_{\uparrow \quad \quad \uparrow} \quad \underbrace{\hspace{1.5cm}}_{y(0) = w_0}$

at $t = 2$. Here we will simply illustrate the steps in the technique when we have $h = 0.5$.

For this problem, $f(t, y) = y - t^2 + 1$; so,

$$w_0 = y(0) = 0.5; \quad w_{i+1} = w_i + h \underbrace{f(t_i, w_i)}_{\substack{t_0 = 0 \\ t_1 = 0 + 0.5 = 0.5 \\ t_2 = 0.5 + 0.5 = 1 \\ t_3 = 1 + 0.5 = 1.5}}$$

$$w_1 = w_0 + 0.5 \left(\underbrace{w_0}_{0.5} - \underbrace{(0.0)^2}_{t_0^2} + 1 \right) = 0.5 + 0.5(1.5) = 1.25;$$

$$w_2 = w_1 + 0.5 \left(w_1 - (0.5)^2 + 1 \right) = 1.25 + 0.5(2.0) = 2.25;$$

$$w_3 = w_2 + 0.5 \left(w_2 - (1.0)^2 + 1 \right) = 2.25 + 0.5(2.25) = 3.375;$$

$$y(2) \approx w_4 = w_3 + 0.5 \left(w_3 - (1.5)^2 + 1 \right) = 3.375 + 0.5(2.125) = 4.4375.$$

EXERCISE SET 5.2

1. Use Euler's method to approximate the solutions for each of the following initial-value problems.

a. $y' = te^{3t} - 2y$, $0 \leq t \leq 1$, $y(0) = 0$, with $h = 0.5$

b. $y' = 1 + (t - y)^2$, $2 \leq t \leq 3$, $y(2) = 1$, with $h = 0.5$

c. $y' = 1 + y/t$, $1 \leq t \leq 2$, $y(1) = 2$, with $h = 0.25$

d. $y' = \cos 2t + \sin 3t$, $0 \leq t \leq 1$, $y(0) = 1$, with $h = 0.25$

3. The actual solutions to the initial-value problems in Exercise 1 are given here. Compare the actual error at each step to the ~~error bound~~.

a. $y(t) = \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}$

b. $y(t) = t + \frac{1}{1-t}$

c. $y(t) = t \ln t + 2t$

d. $y(t) = \frac{1}{2}\sin 2t - \frac{1}{3}\cos 3t + \frac{4}{3}$

1. Use Euler's method to approximate the solutions for each of the following initial-value problems.

a. $y' = te^{3t} - 2y$, $0 \leq t \leq 1$, $y(0) = 0$, with $h = 0.5$

$$w_{i+1} = w_i + h f(t_i, w_i)$$

$$f(t, y) = te^{3t} - 2y$$

$$f(t, w) = te^{3t} - 2w$$

$$f(t_i, w_i) = t_i e^{3t_i} - 2w_i$$

$$w_{i+1} = w_i + h [t_i e^{3t_i} - 2w_i]$$

$$w_1 = w_0 + 0.5 [0 e^{3(0)} - 2w_0] = 0 + 0.5 [0 - 2(0)] = 0 + 0.5 [0] = 0$$

$$w_2 = w_1 + h [t_1 e^{3t_1} - 2w_1] = 0 + 0.5 [0.5 e^{3(0.5)} - 2(0)] = 1.120422$$

$$w_0 = 0$$

t_i	w_i
$t_0 = 0$	$w_0 = 0$
$t_1 = 0.5$	$w_1 = 0$
$t_2 = 1$	$w_2 = 1.120422$

c. $y' = 1 + y/t$, $1 \leq t \leq 2$, $y(1) = 2$, with $h = 0.25$

$$f(t, y) = 1 + \frac{y}{t}$$

$$w_0 = 2$$

$$f(t, w) = 1 + \frac{w}{t}$$

$$f(t_i, w_i) = 1 + \frac{w_i}{t_i}$$

$$w_0 = 2$$

$$w_{i+1} = w_i + h f(t_i, w_i) = w_i + h \left[1 + \frac{w_i}{t_i} \right]$$

$$w_1 = w_0 + 0.25 \left[1 + \frac{w_0}{t_0} \right] = 2 + 0.25 \left[1 + \frac{2}{1} \right] = 2.75$$

$$w_2 = w_1 + 0.25 \left[1 + \frac{w_1}{t_1} \right] = 2.75 + 0.25 \left[1 + \frac{2.75}{1.25} \right] = 3.55$$

$$w_3 = w_2 + 0.25 \left[1 + \frac{w_2}{t_2} \right] = 3.55 + 0.25 \left[1 + \frac{3.55}{1.5} \right] = 4.391667$$

$$w_4 = w_3 + 0.25 \left[1 + \frac{w_3}{t_3} \right] = 4.391667 + 0.25 \left[1 + \frac{4.391667}{1.75} \right] = 5.269048$$

t_i	w_i
$t_0 = 1$	$w_0 = 2$
$t_1 = 1.25$	$w_1 = 2.75$
$t_2 = 1.5$	$w_2 = 3.55$
$t_3 = 1.75$	$w_3 = 4.391667$
$t_4 = 2$	$w_4 = 5.269048$

3) a. $y(t) = \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}$

t_i	u_i (approximate)	y_i (exact value)	Actual error
$t_0 = 0$	$u_0 = 0$	$y_0 = 0$	0
$t_1 = 0.5$	$u_1 = 0$	$y_1 = 0.283617$	0.283617
$t_2 = 1$	$u_2 = 1.120422$	$y_2 = 3.219099$	2.098677

$$y(0) = \frac{1}{5}(0)e^{3(0)} - \frac{1}{25}e^{3(0)} + \frac{1}{25}e^{-2(0)}$$

$$y(0.5) = \frac{1}{5}(0.5)e^{3(0.5)} - \frac{1}{25}e^{3(0.5)} + \frac{1}{25}e^{-2(0.5)}$$

$$y(1) = \frac{1}{5}(1)e^{3(1)} - \frac{1}{25}e^{3(1)} + \frac{1}{25}e^{-2(1)}$$

3)

c. $y(t) = t \ln t + 2t$

$$y(1) = 1 \ln 1 + 2(1) = 2$$

$$y(1.25) = 1.25 \ln 1.25 + 2(1.25) = 2.7789$$

$$y(1.5) = 1.5 \ln 1.5 + 2(1.5) = 3.6081976$$

$$y(1.75) = 1.75 \ln 1.75 + 2(1.75) = 4.4723$$

$$y(2) = 2 \ln 2 + 2(2) = 5.386294$$

t_i	u_i	y_i	Actual error
$t_0 = 1$	$u_0 = 2$	$y_0 = 2$	0
$t_1 = 1.25$	$u_1 = 2.75$	$y_1 = 2.7789$	0.0289
$t_2 = 1.5$	$u_2 = 3.55$	$y_2 = 3.6081976$	0.581976
$t_3 = 1.75$	$u_3 = 4.391667$	$y_3 = 4.4793276$	0.0876606
$t_4 = 2$	$u_4 = 5.269048$	$y_4 = 5.386294$	0.117246